

Intermediate Hands-On Lab – Hard Link & Soft Link

◆ Task 1: Inode Investigation Lab

Objective:

Understand how inode behaves with hard links.

Steps:

1. Create a file:

```
echo "Linux Lab" > lab1.txt
```

2. Create two hard links: lab1.txt lab1_hard1.txt & lab1.txt lab1_hard2.txt
3. Check inode numbers:

? Questions:

- Are all inode numbers the same?
- What is the link count?
- What happens if you edit one file?
- What happens if you delete `lab1.txt`?

◆ Task 2: Link Count Monitoring

Objective:

Understand how Linux manages link count.

Steps:

1. Create a file: count.txt
2. Check link count
3. Create 3 hard links with name count1.txt, count2.txt, count3.txt
4. Check again the link count
5. Delete one hard link:

Questions:

- How does link count change?
 - When will the actual data be deleted?
-

◆ Task 3: Cross Filesystem Restriction

Objective:

Prove that hard links cannot cross filesystems.

Steps:

1. Check mounted filesystems:
2. Identify two different filesystems.
3. Try creating hard link across them:

```
ln /path_from_fs1/file.txt /path_in_fs2/file_link.txt
```

Question:

- What error message do you get?
 - Why does this happen?
-

◆ Task 4: Soft Link Across Filesystems

Objective:

Verify that soft links can cross filesystems.

Steps:

`ln -s /path_from_fs1/file.txt /path_in_fs2/softlink.txt`

Check:

`ls -li`

Questions:

- Are inode numbers same?
 - Does it work successfully?
-

◆ Task 5: Directory Linking

Objective:

Compare directory linking behavior.

Steps:

1. Create directory: project
2. Try to create hard link to directory with name project_hard
3. Now create soft link with name project_soft

? Questions:

- What happens with hard link?
 - Does soft link work?
 - Navigate into soft link and verify content.
-

◆ Task 6: Broken (Dangling) Link Detection

🎯 Objective:

Identify broken soft links.

📌 Steps:

1. Create file:

```
echo "Test File" > original.txt
```
2. Create soft link with name shortcut.txt
3. Delete original.txt
4. Check:

```
ls -l
```

? Questions:

- What happens when you `cat shortcut.txt`?
 - How can you identify broken link?
 - How to fix it?
-

◆ Task 7: Real-World Scenario – Version Management

Objective:

Use soft link to manage versions.

Steps:

Create /tmp/app/v1

Create /tmp/app/v2

```
echo "Version 1" > /tmp/app/v1/app.sh
```

```
echo "Version 2" > /tmp/app/v2/app.sh
```

Create soft link to /tmp/app/v1/app.sh with name /tmp/app/app, which points to current version of app

Check:

```
cat /tmp/app/app
```

Switch version:

```
rm /tmp/app/app
```

```
ln -s /tmp/app/v2/app.sh /tmp/app/app
```

Questions:

- Why is soft link better here?
- What would happen if hard link was used?

◆ Task 8: Data Persistence Challenge

Objective:

Understand data removal mechanism.

Steps:

1. Create file:

```
echo "Important Data" > data.txt
```

2. Create 2 hard links: backup1.txt & backup2.txt to original file data.txt

3. Delete original: data.txt

4. Check:

```
cat backup1.txt
```

5. Delete all hard links.

? Question:

- At what point is data permanently deleted?
-